

# Insu Yun

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Korea Advanced Institute of Science and Technology (KAIST)

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## Research Interests

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System security, software security, binary analysis, fuzzing, and applied cryptography.

## Education

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|-------------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Georgia Institute of Technology</b><br>Ph.D. in Computer Science<br>Advisor: Dr. Taesoo Kim              | Aug. 2015 – Dec. 2020 |
| <b>Korea Advanced Institute of Science and Technology (KAIST)</b><br>B.S. in Computer Science & Mathematics | Sep. 2008 – Feb. 2015 |

## Work Experience

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|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>KAIST</b> , Daejeon, South Korea<br>Associate Professor<br>Assistant Professor                                                                                                    | Feb. 2021 –<br>Mar. 2025 – Present<br>Feb. 2021 – Feb. 2025 |
| <b>Microsoft Research</b> , Research Intern, Seattle, WA<br>Contributed to REPT, a system that utilizes Intel Processor Trace to diagnose production failures<br>Mentor: Weidong Cui | May. 2017 – Aug. 2017                                       |
| <b>Georgia Tech</b> , Research Assistant, Atlanta, GA                                                                                                                                | Aug. 2015 – Dec. 2020                                       |
| <b>Korean Cyber Command</b> , Software Developer, Seoul, Korea<br>Served for the mandatory military service                                                                          | Apr. 2012 – Jan. 2014                                       |

## Publications

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\* Students advised by myself are underlined.

### International Conferences (**Top-tier** and others)

30. **PatchIsland: Orchestration of LLM Agents for Continuous Vulnerability Repair (to appear)**  
Wonyoung Kim\*, Seunggi Min\*, Minjae Gwon, Haein Lee, Dowoo Baik, Hyeon Heo, Minjae Lee, Min Woo Baek, Yonghui Jin, Younggi Park, Yunjae Choi, Taesoo Kim, Sangdon Park, and **Insu Yun**  
Proceedings of the 42nd Annual Computer Security Applications Conference (ACSAC 2026)  
Los Angeles, CA, December 2026
29. **MTEscape: Bypassing Asynchronous Kernel MTE via Conventional Memory Corruption Vulnerabilities (to appear)**  
Kyeongmin Kim, and **Insu Yun**  
Proceedings of the 33rd ACM Conference on Computer and Communications Security (**CCS 2026**)  
The Hague, The Netherlands, November 2026
28. **Prism: A Multi-Team Orchestration of LLM Agents for Automatic Program Repair (to appear)**  
Yunjae Choi, Min Woo Baek, and **Insu Yun**

27. **QueryHouse: Cross-DBMS Differential Testing with LLM and Query Transpilation (to appear)**  
Seoyeon Oh, Juhyun Song, Jaemin Park, Kwanghee Lee, Minae Hyeon, Minji Kim, Sehyuk Ka, Gwangwun Jung, Brian Pak, and **Insu Yun**  
Proceedings of the 31st European Symposium on Research in Computer Security (ESORICS 2026)  
Rome, Italy, September 2026  
**Best Paper Runner-up Award (3 out of 882 submissions)**
26. **RTCon: Context-Adaptive Function-Level Fuzzing for RTOS Kernels**  
Eunkyu Lee, Junyoung Park, and **Insu Yun**  
Proceedings of the 2026 Annual Network and Distributed System Security Symposium (NDSS 2026)  
February 2026
25. **OTABase: Enhancing Over-the-Air Testing to Detect Memory Crashes in Cellular Basebands**  
CheolJun Park, Marc Egli, BeomSeok Oh, Tuan Dinh Hoang, Suhwan Jeong, Martin Crettol, **Insu Yun**, Mathias Payer, and Yongdae Kim  
Proceedings of the 41st Annual Computer Security Applications Conference (ACSAC 2025)  
Austin, TX, December 2025
24. **CROSS-X: Generalized and Stable Cross-Cache Attack on the Linux Kernel**  
Dongok Kim\*, Juhyun Song\*, and **Insu Yun**  
Proceedings of the 32nd ACM Conference on Computer and Communications Security (CCS 2025)  
Taipei, Taiwan, October 2025  
(Acceptance rates: 14%, 316/2278)
23. **Windows plays Jenga: Uncovering Design Weaknesses in Windows File System Security**  
Dong-uk Kim\*, Junyoung Park\*, Sanghak Oh, Hyoungshick Kim, and **Insu Yun**  
Proceedings of the 32nd ACM Conference on Computer and Communications Security (CCS 2025)  
Taipei, Taiwan, October 2025  
(Acceptance rates: 14%, 316/2278)
22. **LLFuzz: An Over-the-Air Dynamic Testing Framework for Cellular Baseband Lower Layers**  
Tuan Dinh Hoang, Taekkyung Oh, CheolJun Park, **Insu Yun**, and Yongdae Kim  
Proceedings of the 34th USENIX Security Symposium (Security 2025)  
Seattle, WA, August 2025  
(Acceptance rates: 17%, 407/2385)
21. **Too Much of a Good Thing: (In-)Security of Mandatory Security Software for Financial Services in South Korea**  
Taisic Yun, Suhwan Jeong, Yonghwa Lee, Seungjoo Kim, Hyoungshick Kim, **Insu Yun**, and Yongdae Kim  
Proceedings of the 34th USENIX Security Symposium (Security 2025)  
Seattle, WA, August 2025  
(Acceptance rates: 17%, 407/2385)
20. **FirmState: Bringing Cellular Protocol States to Shannon Baseband Emulation**  
Suhwan Jeong, Beomseok Oh, Kwangmin Kim, **Insu Yun**, Yongdae Kim, and CheolJun Park  
Proceedings of the 18th ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec 2025)  
Arlington, VA, June 2025
19. **Automated Attack Synthesis for Constant Product Market Makers**  
Sujin Han, Jinseo Kim, Sung-Ju Lee, and **Insu Yun**

Proceedings of the ACM SIGSOFT International Symposium on Software Testing and Analysis 2025 ([ISSTA 2025](#))  
Trondheim, Norway, June 2025  
(Acceptance rates: 19%, 107/550)

18. **Bridging the Gap between Real-World and Formal Binary Lifting through Filtered-Simulation**  
Jihee Park, **Insu Yun**, and Sukyoung Ryu  
Proceedings of the ACM SIGPLAN International Conference on Object-Oriented Programming, Systems, Languages, and Applications 2025 ([OOPSLA 2025](#))  
Singapore, October 2025  
(Acceptance rates: 44%, 67/151)
17. **RGFuzz: Rule-Guided Fuzzer for WebAssembly Runtimes**  
Junyoung Park, Yunho Kim, and **Insu Yun**  
Proceedings of the 46th IEEE Symposium on Security and Privacy ([Oakland 2025](#))  
San Francisco, CA, May 2025  
(Acceptance rates: 15%, 256/1739)
16. **BaseComp: A Comparative Analysis for Integrity Protection in Cellular Baseband Software**  
Eunsoo Kim\*, Min Woo Baek\*, CheolJun Park, Dongkwan Kim, Yongdae Kim, and **Insu Yun**  
Proceedings of the 32nd USENIX Security Symposium ([Security 2023](#))  
Anaheim, CA, August 2023  
(Acceptance rates: 31%, 442/1444)
15. **QueryX: Symbolic Query on Decompiled Code for Finding Bugs in COTS Binaries**  
HyungSeok Han, JeongOh Kyea, Yonghwi Jin, Jinoh Kang, Brian Park, and **Insu Yun**  
Proceedings of the 44th IEEE Symposium on Security and Privacy ([Oakland 2023](#))  
San Francisco, CA, May 2023  
(Acceptance rates: 17%, 196/1147)
14. **Fuzzing@Home: Distributed Fuzzing on Untrusted Heterogeneous Clients**  
Daehee Jang, Ammar Askar, **Insu Yun**, Stephen Tong, Yiqin Cai, and Taesoo Kim  
Proceedings of the 2022 International Symposium on Research in Attacks, Intrusions and Defenses (RAID 2022)  
October 2022
13. **DoLTest: In-depth Downlink Negative Testing Framework for LTE Devices**  
CheolJun Park\*, Sangwook Bae\*, BeomSeok Oh, Jiho Lee, Eunkyu Lee, **Insu Yun**, and Yongdae Kim  
Proceedings of the 31th USENIX Security Symposium ([Security 2022](#))  
Boston, MA, August 2022  
(Acceptance rates: 18%, 256/1414)
12. **HardsHeap: A Universal and Extensible Framework for Evaluating Secure Allocators**  
**Insu Yun**, Woosun Song, Seunggi Min, and Taesoo Kim  
Proceedings of the 28th ACM Conference on Computer and Communications Security ([CCS 2021](#))  
Seoul, South Korea, November 2021  
(Acceptance rates: 22%, 196/880)
11. **Preventing Use-After-Free Attacks with Fast Forward Allocation**  
Brian Wickman, Hong Hu, **Insu Yun**, Daehee Jang, JungWon Lim, Sanidhya Kashyap, and Taesoo Kim  
Proceedings of the 30th USENIX Security Symposium ([Security 2021](#))  
Vancouver, B.C., Canada, August 2021  
(Acceptance rates: 19%, 246/1316)
10. **BaseSpec: Comparative Analysis of Baseband Software and Cellular Specifications for L3 Protocols**

Eunsoo Kim\*, Dongkwan Kim\*, Cheoljun Park, **Insu Yun**, and Yongdae Kim

Proceedings of the 2021 Annual Network and Distributed System Security Symposium (**NDSS 2021**)

February 2021

(Acceptance rates: 15%, 87/578)

9. **Automatic Techniques to Systematically Discover New Heap Exploitation Primitives**

**Insu Yun**, Dhaval Kapil, and Taesoo Kim

Proceedings of the 29th USENIX Security Symposium (**Security 2020**)

Boston, MA, August 2020

(Acceptance rates: 16%, 157/977)

8. **Fuzzing JavaScript Engines with Aspect-preserving Mutation**

Soyeon Park, Wen Xu, **Insu Yun**, Daehee Jang, and Taesoo Kim

Proceedings of the 41st IEEE Symposium on Security and Privacy (**Oakland 2020**)

San Francisco, CA, May 2020

(Acceptance rates: 12%, 104/841)

**Nominated as a finalist in CSAW Best Applied Research Paper Award 2020**

7. **REPT: Reverse Debugging of Failures in Deployed Software**

Weidong Cui, Xinyang Ge, Baris Kasikci, Ben Niu, Upamanyu Sharma, Ruoyu Wang, and **Insu Yun** (alphabetical)

Proceedings of the 13th USENIX Symposium on Operating Systems Design and Implementation (**OSDI 2018**)

Carlsbad, CA, October 2018

(Acceptance rates: 18%, 47/257)

**Jay Lepreau Best Paper Award (3 out of 257 submissions)**

6. **QSYM: A Practical Concolic Execution Engine Tailored for Hybrid Fuzzing**

**Insu Yun**, Sangho Lee, Meng Xu, Yeongjin Jang, and Taesoo Kim

Proceedings of the 27th USENIX Security Symposium (**Security 2018**)

Baltimore, MD, August 2018

(Acceptance rates: 19%, 100/524)

**Distinguished Paper Award (5 out of 524 submissions)**

5. **CAB-Fuzz: Practical Concolic Testing Techniques for COTS Operating Systems**

Su Yong Kim, Sangho Lee, **Insu Yun**, Wen Xu, Byoungyoung Lee, Youngtae Yun, and Taesoo Kim

Proceedings of the 2017 USENIX Annual Technical Conference (ATC 2017)

Santa Clara, CA, July 2017

4. **APISan: Sanitizing API Usages through Semantic Cross-checking**

**Insu Yun**, Changwoo Min, Xujie Si, Yeongjin Jang, Taesoo Kim, and Mayur Naik

Proceedings of the 25th USENIX Security Symposium (**Security 2016**)

Austin, TX, August 2016

(Acceptance rates: 16%, 72/463)

**Nominated as a finalist in CSAW Best Applied Research Paper Award 2016**

3. **HDFI: Hardware-Assisted Data-Flow Isolation**

Chengyu Song, Hyungon Moon, Monjur Alam, **Insu Yun**, Byoungyoung Lee, Taesoo Kim, Wenke Lee, and Yunheung Paek

Proceedings of the 37th IEEE Symposium on Security and Privacy (**Oakland 2016**)

San Jose, CA, May 2016

(Acceptance rates: 13%, 55/413)

2. **Analyzing Security of Korean USIM-based PKI Certificate Service**

Shinjo Park, Suwan Park, **Insu Yun**, Dongkwan Kim, and Yongdae Kim

Proceedings of the 15th International Workshop on Information Security Applications (WISA 2014)

Jeju Island, Korea, August 2014

## 1. **Kargus: A Highly-scalable Software-based Intrusion Detection System**

Muhammad Jamshed, Jihyung Lee, Sangwoo Moon, **Insu Yun**, Deokjin Kim, Sungryoul Lee, Yung Yi, and KyoungSoo Park

Proceedings of the 19th ACM Conference on Computer and Communications Security (**CCS 2012**)

Raleigh, NC, October 2012

(Acceptance rates: 19%, 81/426)

## International Journal

### 1. **Scalable and Secure Virtualization of HSM with ScaleTrust**

Juhyeng Han, **Insu Yun**, Seongmin Kim, Taesoo Kim, Soeul Son, and Dongsu Han

IEEE/ACM Transactions on Networking (ToN)

November 2022

## Other Refereed Materials

### 5. **CTFusion: A CTF-based Benchmark for LLM Agent Evaluation**

Dongjun Lee, Ga-eun Bae, and **Insu Yun**

Proceedings of the Second Workshop on Agents in the Wild: Safety, Security, and Beyond (ICML AIWILD 2026)

July 2026

### 4. **From the Vulnerability to the Victory: A Chrome Renderer 1-Day Exploit's Journey to v8CTF Glory**

Haein Lee, and **Insu Yun**

TyphoonCon 2024

Seoul, Korea, May 2024

### 3. **One shot, Triple kill: Pwning all three Google kernelCTF instances with a single 1-day Linux vulnerability**

Dongok Kim, Seunghyun Lee, and **Insu Yun**

POC 2023

Seoul, Korea, November 2023

### 2. **Compromising the macOS kernel through Safari by chaining six vulnerabilities**

Yonghwi Jin, Jungwon Lim, **Insu Yun**, and Taesoo Kim

Black Hat USA Briefings (Black Hat USA 2020)

Las Vegas, NV, August 2020

### 1. **AVPASS: Leaking and Bypassing Antivirus Detection Model Automatically**

Jinho Jung, Chanil Jeon, Max Wolotsky, **Insu Yun**, and Taesoo Kim

Black Hat USA Briefings (Black Hat USA 2017)

Las Vegas, NV, July 2017

## Thesis

### 1. **Concolic Execution Tailored for Hybrid Fuzzing**

**Insu Yun**

Ph.D. thesis, Georgia Institute of Technology

Atlanta, GA, December 2020

## Professional Activities

### Technical Program Committee (International)

Program Committee, *IEEE Symposium on Security and Privacy (Oakland)*, 2024–2026

Program Committee, *Network and Distributed System Security Symposium (NDSS)*, 2024–2026

Program Committee, *The International Conference on Cryptology and Network Security (CANS)*, 2025

Program Committee, *ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)*, 2022–2023

### Journal Editor (International)

Associate Editor, *ACM Transaction on Storage (ToS)*, 2024–2026

### Others (International)

Organization Committee (Publicity Chair), *Network and Distributed System Security Symposium (NDSS)*, 2027

Organization Committee (Web Chair), *ACM Conference on Security and Privacy in Wireless and Mobile Networks (WiSec)*, 2024

Artifact Evaluation Committee, *ACM Conference on Computer and Communications Security (CCS)*, 2023

Artifact Evaluation Committee, *USENIX Security Symposium (Security)*, 2023

Organization Committee (Web Chair), *ACM Conference on Computer and Communications Security (CCS)*, 2021

### Technical Program Committee (Domestic)

Program Committee, *Conference on Information Security and Cryptography Summer (CISC-S)*, 2026

Program Committee, *Conference on Information Security and Cryptography Winter (CISC-W)*, 2025

Program Committee, *Conference on Information Security and Cryptography Summer (CISC-S)*, 2021

### Others (Domestic)

Policy Advisory Committee, *Ministry of National Defense*, 2026

Security Special Committee, *National Artificial Intelligence Strategy Committee*, 2026

Security Task Force, *National Artificial Intelligence Strategy Committee*, 2025

Advisory Boards, *AI Cyber Defense Contest (ACDC)*, 2025

Advisory Boards, *HackTheon Sejong*, 2024–2025

Organization Committee, *Conference on Information Security and Cryptography Summer (CISC-S)*, 2021

## Teaching Experience

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|                                                                             |             |
|-----------------------------------------------------------------------------|-------------|
| Advanced Programming Techniques for Electrical Engineering (EE309 at KAIST) | Fall 2025   |
| • Evaluation – Average: 4.29 / 5                                            |             |
| Software Hacking Theory and Practice (EE517 at KAIST)                       | Spring 2025 |
| • Evaluation – Average: 4.67 / 5                                            |             |
| Advanced Programming Techniques for Electrical Engineering (EE309 at KAIST) | Fall 2024   |
| • Evaluation – Average: 4.54 / 5                                            |             |
| Information Security Laboratory (IS521 at KAIST)                            | Fall 2024   |
| • Evaluation – Average: 4.64 / 5                                            |             |
| Software Hacking Theory and Practice (EE517 at KAIST)                       | Spring 2024 |
| • Evaluation – Average: 4.92 / 5                                            |             |
| Advanced Programming Techniques for Electrical Engineering (EE309 at KAIST) | Fall 2023   |
| • Evaluation – Average: 4.57 / 5                                            |             |
| Software Hacking Theory and Practice (EE517 at KAIST)                       | Spring 2023 |
| • Evaluation – Average: 4.54 / 5                                            |             |
| Programming Structures for Electronical Engineering (EE209 at KAIST)        | Fall 2022   |
| • Evaluation – Average: 4.65 / 5                                            |             |
| Software development environment and tools practice (EE485-A at KAIST)      | Fall 2022   |
| • Evaluation – Average: 4.43 / 5                                            |             |
| Software Security (EE595-B at KAIST)                                        | Spring 2022 |
| • Evaluation – Average: 5 / 5                                               |             |
| Programming Structures for Electronical Engineering (EE209 at KAIST)        | Fall 2021   |
| • Evaluation – Average: 4.34 / 5                                            |             |

Software development environment and tools practice (EE485-A at KAIST)

Fall 2021

- Evaluation – Average: 4.34 / 5

Software Security (EE595-B at KAIST)

Spring 2021

- Evaluation – Average: 4.9 / 5

## Honors & Awards

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### Academic awards

|                                                                                      |            |
|--------------------------------------------------------------------------------------|------------|
| Best Paper Runner-up Award, ESORICS 2026 (3 out of 882 submissions)                  | Sep. 2026  |
| KAIST Q-Day Special Commendation for Research, KAIST                                 | Nov. 2025  |
| Song-Am Future Researcher Award, KAIST                                               | Feb. 2025  |
| Technology Innovation Awards, KAIST College of Engineering                           | Dec. 2024  |
| Best Teaching Award, KAIST Electrical Engineering                                    | Oct. 2024  |
| Prize for Excellence in Teaching, KAIST                                              | Feb. 2024  |
| Frontiers of Science Award, The First International Congress of Basic Science (ICBS) | July. 2023 |
| Best Teaching Award, KAIST Electrical Engineering                                    | Sep. 2021  |
| Jay Lepreau Best Paper Award, USENIX OSDI 2018                                       | Aug. 2018  |
| Distinguished Paper Award, USENIX Security 2018                                      | Aug. 2018  |

### Hacking competitions

|                                                |           |
|------------------------------------------------|-----------|
| DEFCON 34 CTF, 2nd place (Team 0x4b52)         | Aug. 2026 |
| DEFCON 26 CTF, 1st place (Team DEFKOR00T)      | Aug. 2018 |
| DEFCON 24 CTF, 3rd place (Team DEFKOR)         | Aug. 2016 |
| DEFCON 23 CTF, 1st place (Team DEFKOR)         | Aug. 2015 |
| Whitehat contest 2014, 1st place (Team SysSec) | Nov. 2014 |
| DEFCON 22 CTF, 10th place (Team GoN)           | Aug. 2014 |
| SECCON CTF 2014, 1st place (TOEFL Beginner)    | Feb. 2014 |
| Codegate CTF 2012, 3rd place (Team GoN)        | Apr. 2012 |
| Secuinside CTF, 3rd place (Team GoN)           | Oct. 2011 |
| ISEC CTF, 1st place (Team GoN)                 | Sep. 2011 |
| DEFCON 18 CTF, 3rd place (Team GoN)            | Aug. 2010 |
| Codegate CTF 2010, 5th place (Team GoN)        | Apr. 2010 |
| KISA HDCON, Gold Medal, 2nd place (Team GoN)   | May 2009  |
| Codegate CTF 2009, 4th place (Team GoN)        | Apr. 2009 |

### Scholarships

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|---------------------------------------------------------------------|-----------------------|
| National Research Foundation of Korea Scholarship for Undergraduate | Mar. 2008 – Dec. 2013 |
|---------------------------------------------------------------------|-----------------------|

### Others

|                                                                              |           |
|------------------------------------------------------------------------------|-----------|
| DARPA AIxCC, 1st place (Team Atlanta), \$4M award                            | Aug. 2025 |
| Cyber Security Challenge, 2nd place (Team HackingLab), \$400K research grant | 2023      |
| DARPA Cyber Grand Challenge, Finalist (Team Disekt)                          | Aug. 2016 |
| KISA Bug Bounty Program's Hall of Fame                                       | 2013      |

## Vulnerability Discovery Reward (aka Bug bounty)

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To summarize, \$244.6K (by my students) and \$92.8K (by myself) bug bounties are awarded so far.

### By my students

|                                                                           |           |
|---------------------------------------------------------------------------|-----------|
| Pwn2Own - Microsoft Edge and Google Chrome (\$145K), ZDI, by SeunHyun Lee | Mar. 2024 |
| v8CTF - CVE-2023-6702 (\$10K), Google, by Haein Lee                       | Jan. 2024 |

|                                                                                                     |           |
|-----------------------------------------------------------------------------------------------------|-----------|
| kernelCTF - CVE-2023-3390 (\$67.8K), Google, by Dongok Kim and SeunHyun Lee                         | Oct. 2023 |
| Type confusion in V8 (\$7K), Google, by Haein Lee                                                   | Mar. 2023 |
| NAS authentication bypass in Exynos (\$14.8K), Samsung Electronics, by Eunsoo Kim and CheolJun Park | Feb. 2022 |

### By myself

|                                                                                                        |           |
|--------------------------------------------------------------------------------------------------------|-----------|
| PSV-2021-0304: afpd auth bypass (\$300), NETGEAR                                                       | Mar. 2021 |
| Pwn2Own Apple Safari with a kernel privilege escalation (\$70K), ZDI, with Yonghwi Jin and Jungwon Lim | Mar. 2020 |
| Apple Safari sandbox escape (\$20K), Apple                                                             | Dec. 2019 |
| Three integer overflow vulnerabilities in PHP (\$1.5K), the Internet Bug Bounty                        | Jun. 2016 |
| An Integer Overflow in Python zipimport (\$1K), the Internet Bug Bounty                                | Apr. 2016 |

## Patents

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### International

#### 3. Dynamic testing method and system for detecting memory corruptions in lower layers of cellular baseband of mobile communication devices (Pending)

Inventors: Yongdae Kim, Hoang Dinh Tuan, Insu Yun, Taekkyung Oh, Beomseok Oh, Mincheol Son, CheolJun Park

Application date: 2026.01.07

Application number: 19442669

Country: US

#### 2. Security analysis system and method based on negative testing for protocol implementation of LTE device

Inventors: Yongdae Kim, CheolJun Park, Sangwook Bae, Beomseok Oh, Jiho Lee, Mincheol Son, Insu Yun

Registration date: 12511402

Patent number: 2025.12.30

Country: US

#### 1. Reverse debugging of software failures

Inventors: Weidong Cui, Xinyang Ge, Baris Kasikci, Cengiz Can, Ben Niu, Ruoyu Wang, Insu Yun

Registration date: 10565511

Patent number: 2020.02.18

Country: US

### Domestic

#### 9. Method and apparatus for detecting memory vulnerabilities based on network-side state control in wireless communication systems (Pending)

Inventors: Yongdae Kim, Beomseok Oh, Insu Yun, Mincheol Son, Hoang Dinh Tuan, Soohwan Jeong, CheolJun Park

Application date: 2025.12.03

Application number: 10-2025-0189113

Country: Korea

#### 8. Method and apparatus for fuzzing-based vulnerability detection (Pending)

Inventors: Insu Yun, Eunkyoo Lee, Junyeong Park

Application date: 2025.11.11

Application number: 10-2025-0169729

Country: Korea

**7. Dynamic testing method and system for detecting memory vulnerabilities in lower layers of cellular baseband of mobile communication devices (Pending)**

Inventors: Yongdae Kim, Taekkyung Oh, Beomseok Oh, Mincheol Son, CheolJun Park, Hoang Dinh Tuan, Insu Yun

Application date: 2025.05.07

Application number: 10-2025-0059118

Country: Korea

**6. Heuristic-based binary lifting and verification system (Pending)**

Inventors: Sukyoung Ryu, Jihee Park, Insu Yun

Application date: 2025.03.31

Application number: 10-2025-0041131

Country: Korea

**5. Method and apparatus for decompilation-specialized static binary translation**

Inventors: Insu Yun, Woosun Song, Seunggi Min

Registration date: 10-2834292-0000

Patent number: 2025.07.10

Country: Korea

**4. Security analysis system and method based on negative testing for protocol implementation of LTE device**

Inventors: Yongdae Kim, CheolJun Park, Sangwook Bae, Beomseok Oh, Jiho Lee, Mincheol Son, Insu Yun

Registration date: 10-2514797-0000

Patent number: 2023.03.23

Country: Korea

**3. Method and system for automatically analyzing bugs in cellular baseband software using comparative analysis based on cellular specifications**

Inventors: Yongdae Kim, Eunsoo Kim, Dongkwan Kim, CheolJun Park, Insu Yun

Registration date: 10-2546946-0000

Patent number: 2023.06.20

Country: Korea

**2. Methods and systems for key management service provision**

Inventors: Dongsoo Han, JuHyeng Han, Insu Yun

Registration date: 10-2803138-0000

Patent number: 2025.04.28

Country: Korea

**1. Security analysis system based on negative testing for protocol implementation of LTE device (Pending)**

Inventors: Yongdae Kim, CheolJun Park, Sangwook Bae, Beomseok Oh, Jiho Lee, Mincheol Son, Insu Yun

Application date: 2021.10.07

Application number: 10-2021-0133149

Country: Korea

# Invited Talks

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## International

- Title: Atlantis: an Autonomous LLM-Powered Bug Finding and Fixing System  
– Presented at KAIST-NYU Workshop Nov. 2025
- Title: How to build Skynet — a system that hacks systems  
– **Keynote** speech at TyphoonCon, Seoul, Korea Jun. 2023
- Title: HardsHeap: A Universal and Extensible Framework for Evaluating Secure Allocators  
– Presented at ACM CCS 2021, Online Nov. 2021
- Title: Automatic Techniques to Systematically Discover New Heap Exploitation Primitives  
– Presented at USENIX Security 2020, Online Aug. 2020
- Title: QSYM: A Practical Concolic Execution Engine Tailored for Hybrid Fuzzing  
– Presented at USENIX Security 2018, Baltimore, MD Aug. 2018
- Title: APISan: Sanitizing API Usages through Semantic Cross-checking  
– Presented at USENIX Security 2016, Austin, TX Aug. 2016

## Domestic

- Title: The Singularity is Here: When AI Meets Cyber Security  
– Seminar at Tech-Share by Samsung Electronics, Hyundai Motor, and Naver, Seoul, Korea Feb. 2026
- Title: The Dark and Bright Sides of AI in Cybersecurity  
– Seminar at Samsung Electronics DS, Seoul, Korea Jan. 2026  
– Seminar at AI Security Technology Seminar, Seoul, Korea Dec. 2025  
– Seminar at KAIST AI Graduate School Colloquium Oct. 2025  
– Seminar at KISA, Seoul, Korea Oct. 2025  
– Seminar at KISIA, Seoul, Korea Oct. 2025  
– Seminar at KAIST-Cyber Security Policy Program Sep. 2025  
– Seminar at Samsung SDS, Seoul, Korea Sep. 2025  
– Seminar at National Intelligence Service, Seoul, Korea Sep. 2025
- Title: AI that Changed the Cybersecurity Talent and Education  
– **Keynote** at Cybersecurity Talent Development Symposium, Seoul, Korea Dec. 2025  
– Seminar at Cyber Security Policy Forum, Seoul, Korea Dec. 2025
- Title: The Future of Security Transformed by AI  
– Seminar at ACDC (AI Cyber Defense Contest), Seoul, Korea Dec. 2025
- Title: AI that Changed the Battlefield  
– **Keynote** at WhiteHat 2025, Seoul, Korea Nov. 2025
- Title: Atlantis: an Autonomous LLM-Powered Bug Finding and Fixing System  
– Seminar at Cyber Security Society, Seoul, Korea Nov. 2025  
– Seminar at Cyber Summit Korea 2025, Seoul, Korea Sep. 2025  
– Seminar at Tech-Share by Samsung Electronics, Hyundai Motor, and Naver, Seoul, Korea Aug. 2025  
– Seminar at ENKI WhiteHat, Seoul, Korea Sep. 2025  
– Seminar at LIGNex1, Seoul, Korea Sep. 2025  
– Seminar at KCSCON, Seoul, Korea Sep. 2025
- Title: AIxCC: The Future of Cybersecurity with LLMs  
– Seminar at ETRI, Daejeon, Korea Nov. 2024  
– Seminar at Moon Sool Graduate School of Future Strategy, Daejeon, Korea Nov. 2024  
– Seminar at Future Security Tech Forum, Jeju, Korea Nov. 2024  
– Seminar at National Security Research Institute (NSRI), Daejeon, Korea Sep. 2024
- Title: 2024 Security Strategy: Polarization  
– Seminar at Defense Counterintelligence Command, Gwacheon, Korea May. 2024
- Title: Building Automated Hacking Systems  
– Seminar at POSTECH, Pohang, Korea Nov. 2023

|                                                                                                          |           |
|----------------------------------------------------------------------------------------------------------|-----------|
| Title: Trends in Security Vulnerabilities of Low Earth Orbit Satellites                                  |           |
| – Presented at ETRI, Daejeon, Korea                                                                      | Aug. 2023 |
| Title: Academic Research from Offensive Research                                                         |           |
| – Presented at Samsung, Seoul, Korea                                                                     | Aug. 2023 |
| Title: Human-friendly binary analysis                                                                    |           |
| – Presented at ETRI, Daejeon, Korea                                                                      | Nov. 2023 |
| – Presented at Korea Computer Congress (KCC), Seoul, Korea                                               | Jun. 2023 |
| Title: Exploit in the wild                                                                               |           |
| – Presented at ETRI, Daejeon                                                                             | Jun. 2023 |
| Title: Hacking 101                                                                                       |           |
| – Presented at WISC, Seoul                                                                               | Sep. 2022 |
| Title: Attack and Defenses for Heap Vulnerabilities in 2022                                              |           |
| – Seminar at ETRI, Daejeon                                                                               | Apr. 2022 |
| Title: Comparative Analysis of Baseband Software and Cellular Specifications for Finding Vulnerabilities |           |
| – Seminar at UNIST, Ulsan                                                                                | May. 2023 |
| – Seminar at Security@KAIST, Online                                                                      | Jun. 2022 |
| – Seminar at Cyber Operations Command, Seoul                                                             | Jun. 2022 |
| Title: Scalable and Automatic Vulnerability Discovery Beyond Random Testing                              |           |
| Seminar at Seoul National University, Seoul, Korea, Mar. 2019                                            |           |
| Title: Memory Allocator Security                                                                         |           |
| – Presented at Best of Best (BoB), Seoul                                                                 | Feb. 2023 |
| – Presented at Computer System Society Conference (CSC), Pyeongchang                                     | Feb. 2023 |
| – Seminar at UNIST, Online                                                                               | May. 2022 |
| – Seminar at Yonsei university, Online                                                                   | Apr. 2022 |
| – Seminar at Sungkyunkwan university, Online                                                             | Apr. 2022 |
| – Seminar at ETRI, Daejeon                                                                               | Jan. 2022 |
| – Seminar at National Security Research Institute (NSRI), Daejeon                                        | Dec. 2021 |
| – Seminar at Securty@KAIST, Online                                                                       | Nov. 2021 |
| – Seminar at KAIST GSIS, Online                                                                          | Nov. 2021 |
| Title: Browser Security: Hacking & Research                                                              |           |
| – Presented at Open Theori Research Seminar #6, Online                                                   | Dec. 2021 |
| – Seminar at Hanyang University, Online                                                                  | Nov. 2021 |
| – Presented at KR Becks Meetup #1 by LINE, Online                                                        | Aug. 2021 |
| – Seminar at Security@KAIST, Online                                                                      | Jun. 2021 |

## Grants

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To summarize, 2.4B KRW is awarded, and my portion is 1.96B KRW.

- |                                                                                                       |               |
|-------------------------------------------------------------------------------------------------------|---------------|
| 19. <b>Intelligent multi-agent system for variant vulnerability detection in large-scale software</b> | 26.03 – 27.02 |
| Agency/Company: NRF                                                                                   |               |
| Money: 142M KRW                                                                                       |               |
| Role: PI                                                                                              |               |
| 18. <b>Design and implementation of novel fuzzing system for USB driver security</b>                  | 25.08 – 26.05 |
| Agency/Company: Samsung Electronics                                                                   |               |
| Money: 82.5M KRW                                                                                      |               |
| Role: PI                                                                                              |               |

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|-------------------------------------------------------------------------------------------------------------------------------|---------------|
| 17. <b>SAT-HACK: Satellite cyber kill chain verification research</b>                                                         | 25.07 – 27.02 |
| Agency/Company: LIGNex1                                                                                                       |               |
| Money: 275M KRW                                                                                                               |               |
| Role: PI                                                                                                                      |               |
| 16. <b>Software safety research based on automated vulnerability pattern generation</b>                                       | 25.04 – 25.10 |
| Agency/Company: NSRI                                                                                                          |               |
| Money: 60M KRW                                                                                                                |               |
| Role: PI                                                                                                                      |               |
| 15. <b>AIxCC: Automating real-world vulnerability discovery and remediation</b>                                               | 24.09 – 26.02 |
| Agency/Company: NYU                                                                                                           |               |
| Money: 99.9M KRW                                                                                                              |               |
| Role: PI                                                                                                                      |               |
| 14. <b>Research on building an open source kernel security model</b>                                                          | 24.04 – 24.10 |
| Agency/Company: NSRI                                                                                                          |               |
| Money: 54.5M KRW                                                                                                              |               |
| Role: PI                                                                                                                      |               |
| 13. <b>Revisiting IoT threat models for smart cities and developing a vulnerability analysis system based on these models</b> | 24.01 – 25.12 |
| Agency/Company: IITP                                                                                                          |               |
| Money: 400M KRW                                                                                                               |               |
| Role: PI                                                                                                                      |               |
| 12. <b>Building a system to assist variant analysis for browsers</b>                                                          | 23.06 – 24.05 |
| Agency/Company: NRF                                                                                                           |               |
| Money: 65M KRW                                                                                                                |               |
| Role: PI                                                                                                                      |               |
| 11. <b>Generating a security model based on JavaScript intermediate language</b>                                              | 23.04 – 23.10 |
| Agency/Company: NSRI                                                                                                          |               |
| Money: 54.5M KRW                                                                                                              |               |
| Role: PI                                                                                                                      |               |
| 10. <b>Verifying security threats in open-source operating systems</b>                                                        | 23.04 – 23.10 |
| Agency/Company: NSRI                                                                                                          |               |
| Money: 54.5M KRW                                                                                                              |               |
| Role: PI                                                                                                                      |               |
| 9. <b>Browser fuzzing with formal verification for cross architectures</b>                                                    | 22.09 – 23.09 |
| Agency/Company: NRF                                                                                                           |               |
| Money: 110M KRW                                                                                                               |               |
| Role: PI                                                                                                                      |               |
| 8. <b>Building test suites for validating vulnerability detection</b>                                                         | 22.08 – 22.11 |
| Agency/Company: ETRI                                                                                                          |               |
| Money: 27.3M KRW                                                                                                              |               |
| Role: PI                                                                                                                      |               |
| 7. <b>Generating a security model based on JavaScript security analysis</b>                                                   | 22.04 – 22.10 |
| Agency/Company: NSRI                                                                                                          |               |

- Money: 54.5M KRW  
Role: PI
6. **Developing techniques for collection and integrated analysis of automotive systems through event-based experimental systems** 22.04 – 23.12  
Agency/Company: Dankook university  
Money: 300M  $\times$  0.5 KRW  
Role: Co-PI with Prof. Yujip Won
  5. **6G security** 21.08 – 23.09  
Agency/Company: Samsung Electronics  
Money: 200M  $\times$  0.2 KRW  
Role: Co-PI with Prof. Yongdae Kim
  4. **DRAM security** 21.07 – 24.06  
Agency/Company: Samsung Electronics  
Money: 180M  $\times$  0.2 KRW  
Role: Co-PI with Prof. Yongdae Kim
  3. **Systematic and precise transformation of the Qualcomm Hexagon architecture into intermediate representations for binary analysis** 21.06 – 22.05  
Agency/Company: NRF  
Money: 46.7M KRW  
Role: PI
  2. **Automatically generating a security model for discovering web browser vulnerabilities** 21.04 – 21.10  
Agency/Company: NSRI  
Money: 54.5M KRW  
Role: PI
  1. **Developing a scalable cyber reasoning system (Start-up)** 21.02 – 24.12  
Agency/Company: KAIST  
Money: 150M KRW  
Role: PI

## Advising and Mentoring

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### Ph.D. Students

- Youngmin Choi (Co-advising with Prof. Kyoungsoo Park)
- Haein Lee
- Sujin Han
- Junyoung Park
- Minwoo Baek
- Eunkyu Lee
- Donghyeon Kim
- Dongjun Lee

### Master Students

- Kyeongmin Kim

- Juhyun Song
- Minwoo Jeong
- Gunhee Ahn

### **Alumni (Ph.D.)**

- HyungSeok Han (Co-advised with Yongdae Kim), Postdoc at Georgia Tech Ph.D. in 2023

### **Alumni (M.S.)**

- Dong-uk Kim, 78ResearchLab M.S. in 2025
- Dongok Kim, ENKI WhiteHat M.S. in 2025
- Hyeon Heo, ENKI WhiteHat M.S. in 2025
- Seunggi Min, Financial Security Institute M.S. in 2025
- Wonyoung Kim, Samsung M.S. in 2025
- Taisic Yun (Co-advised with Yongdae Kim), Theori M.S. in 2024
- Wonyoung Jung, 78ResearchLab M.S. in 2024
- Hyunsik Jeong (Co-advised with Yongdae Kim), S2W M.S. in 2021